CYBERSECURITY INVESTIGATION REPORT

DNS Investigation Using Wireshark

Mission 014 – Packet Capture and Analysis of DNS Traffic

Report title	DNS Investigation Using Wireshark
Investigation	Mission 014 – DNS Investigation
Prepared by	Huzaifa Saeed
Tools used	Wireshark (packet analysis); nslookup and ipconfig (Windows command line)
Operating system	Microsoft Windows
Evidence file	DNS-Investigation.pcapng
Scope	Observation and analysis of DNS queries and responses on a local network; no malicious activity was concluded

1. Introduction

This investigation examined Domain Name System (DNS) traffic using Wireshark and Windows command-line utilities. The objective was to observe DNS queries and responses, identify the DNS resolver, inspect packet-level details, and investigate common DNS record types.

DNS translates human-readable hostnames into IP addresses and underpins nearly all network communication, which makes it a core data source for security analysts reviewing network behaviour. This report documents the traffic that was generated and captured, the analysis performed at packet level, and the findings drawn from the evidence.

1.1 Objectives

- Observe DNS queries and responses generated from a Windows client.
- Identify the DNS resolver serving the client.
- Inspect DNS query and response packets at the field level (addresses, ports, transaction ID, status, answer data and TTL).
- Investigate and document the A, AAAA, MX, NS, TXT, CNAME and PTR record types.

1.2 Summary of Key Observations

DNS client (IPv4)	192.168.1.31
DNS resolver (IPv4)	192.168.1.1
DNS resolver (IPv6 link-local)	fe80::1 (as reported by nslookup)
Resolver hostname	gpon.net (reported by nslookup and confirmed by reverse lookup)
Primary transport	UDP, destination port 53
Detailed query/response pair	Packets 236 (query) and 237 (response) – clients4.google.com, type A
Record types identified	A, AAAA, MX, NS, TXT, CNAME, PTR
Malicious activity	None concluded from the observed traffic

2. Lab Environment and Methodology

2.1 Lab Environment

Component	Detail
Operating system	Microsoft Windows
Packet analysis tool	Wireshark
Command-line utility	nslookup
Network connection	Local network with internet access
DNS client IPv4 address (observed in capture)	192.168.1.31
DNS resolver IPv4 address (observed in capture)	192.168.1.1
DNS resolver IPv6 link-local address (reported by nslookup)	fe80::1
Resolver hostname (nslookup, confirmed by reverse lookup)	gpon.net

2.2 Methodology

The investigation followed these steps:

- **Clear the cache.** The Windows DNS resolver cache was flushed with ipconfig /flushdns so that subsequent lookups would generate fresh network queries.
- **Generate DNS traffic.** Five domains were queried with nslookup: google.com, microsoft.com, youtube.com, github.com and cloudflare.com.
- **Capture and filter.** DNS traffic was captured in Wireshark and reviewed using the dns display filter.
- **Inspect packets.** A query/response pair (packets 236 and 237) was expanded and examined field by field.
- **Investigate record types.** A, AAAA, MX, NS, TXT, CNAME and PTR queries were issued and their responses examined in Wireshark.
- **Preserve evidence.** Screenshots were captured and the packet capture was saved as DNS-Investigation.pcapng.

3. DNS Cache and Domain Resolution

The Windows DNS resolver cache was cleared before testing by running `ipconfig /flushdns`. The five domains were then queried using nslookup. The lookups completed successfully and returned IPv4 and, where available, IPv6 addresses.

Domain	IPv4 address(es)	IPv6 address(es)
google.com	142.250.200.174	2a00:1450:4019:80f::200e
microsoft.com	150.171.109.163	2603:1061:14:a2::1
youtube.com	142.251.37.142	2a00:1450:4019:819::200e
github.com	20.207.73.82	–
cloudflare.com	104.16.132.229, 104.16.133.229	2606:4700::6810:84e5, 2606:4700::6810:85e5

DNS answers can vary over time and between resolvers because services may use multiple addresses and distributed infrastructure.

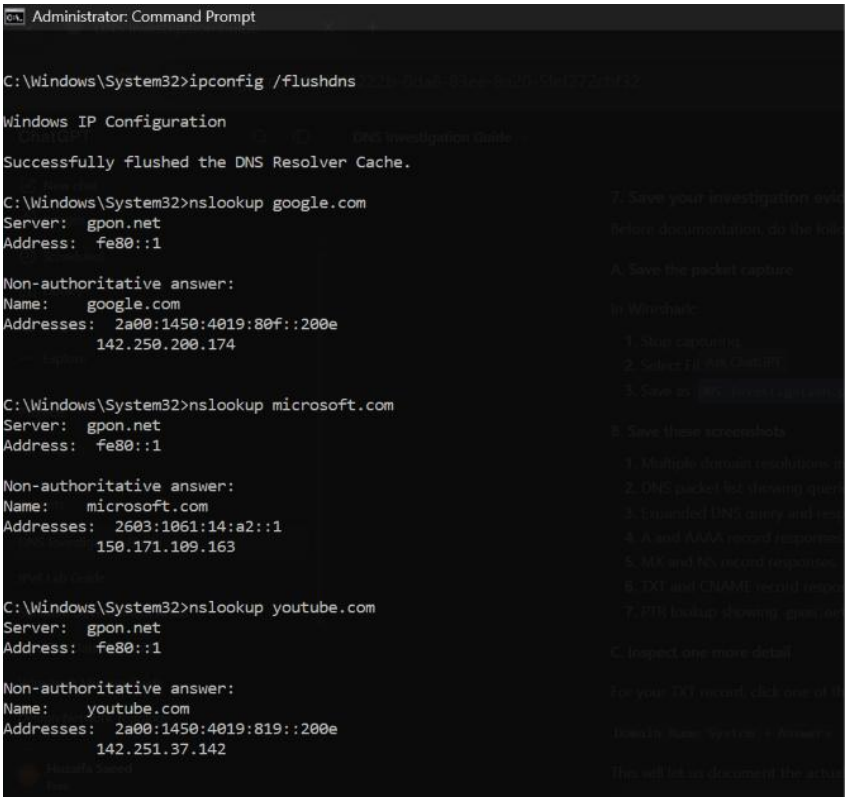


Figure 1 – *Cache flush (ipconfig /flushdns) followed by nslookup results for google.com, microsoft.com and youtube.com. Each response reports Server: gpon.net, Address: fe80::1 and a non-authoritative answer. The screenshot shows the first three lookups; github.com and cloudflare.com results are recorded in the table above.*

4. DNS Server Identification

The nslookup output identified the DNS server as:

Server	gpon.net
Address	fe80::1

The Wireshark capture also showed DNS traffic between the Windows client (192.168.1.31) and the local resolver (192.168.1.1). IPv6 DNS exchanges were also visible between the client's link-local IPv6 address and fe80::1.

These observations indicate that the local router/resolver provides DNS service to the client.

> **Scope note:** The capture identifies the local resolver; it does not establish which upstream recursive resolver, if any, the router uses.

5. Packet Analysis

5.1 DNS Traffic Overview

Applying the dns display filter in Wireshark isolated the DNS traffic in the capture. Both IPv4 exchanges (192.168.1.31 and 192.168.1.1) and IPv6 link-local exchanges (fe80::1) are present. Packets 236 and 237, selected for detailed analysis, appear in the list below.

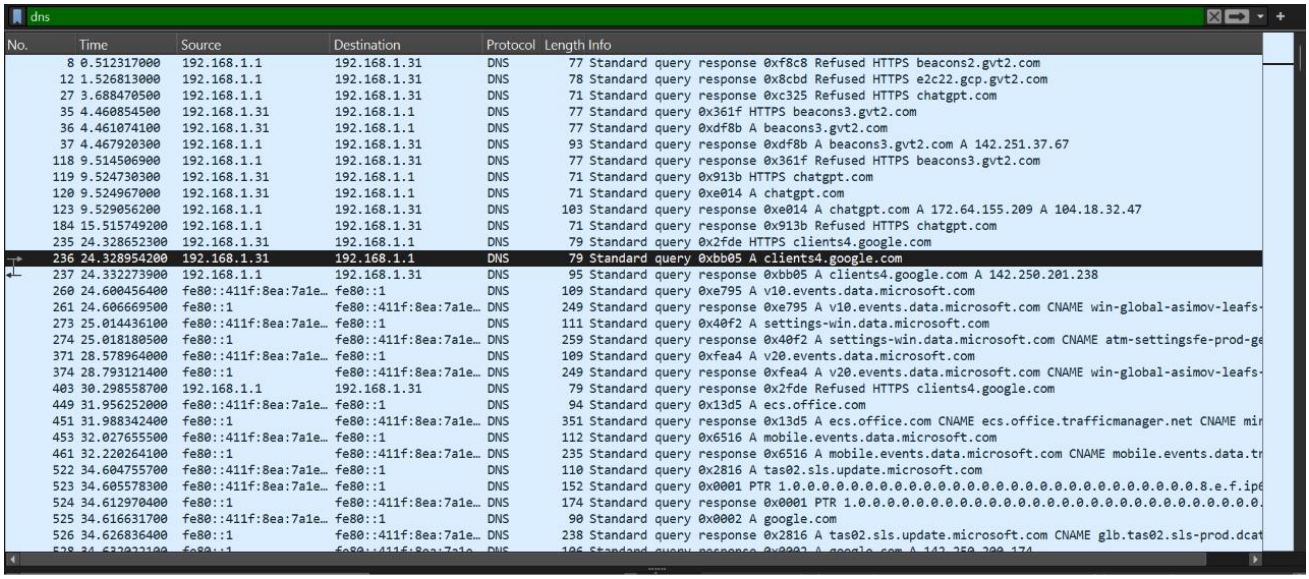


Figure 2 – *Wireshark packet list filtered with "dns". Packet 236 (query 0xbb05, A clients4.google.com) is highlighted, followed by its response in packet 237 (A 142.250.201.238).*

5.2 Query Packet 236

Packet 236 – DNS Query

Client / source IPv4: 192.168.1.31
Resolver / destination IPv4: 192.168.1.1
Transport protocol: UDP
Client source port: 59829
DNS destination port: 53
Transaction ID: 0xbb05
Requested name: clients4.google.com
Requested record type: A
Also visible in screenshot: Frame length 79 bytes; IP TTL 128; flags 0x0100 (standard query); 1 question; “Response In: 237”

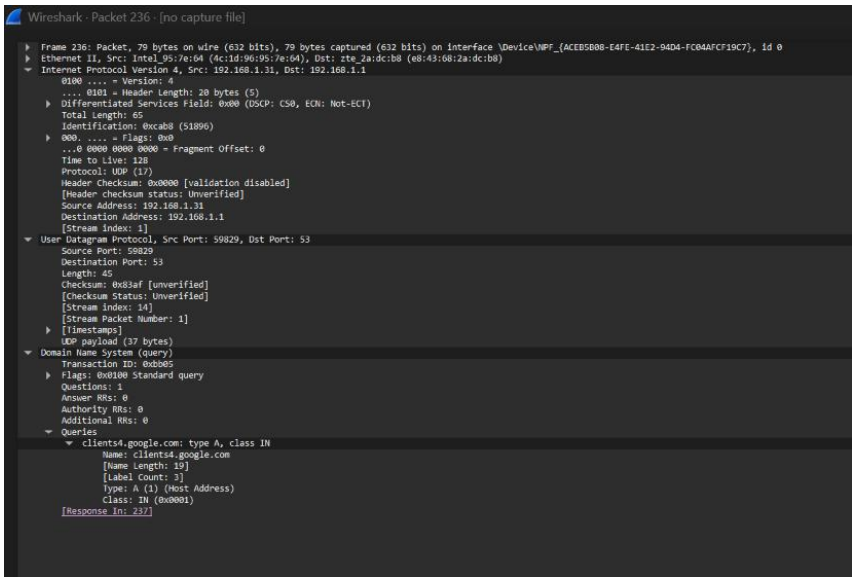


Figure 3 – *Expanded query packet 236 showing the IPv4, UDP and DNS layers for the A query to clients4.google.com.*

5.3 Response Packet 237

Packet 237 – DNS Response

Resolver / source IPv4: 192.168.1.1
Client / destination IPv4: 192.168.1.31
UDP source port: 53
Client destination port: 59829
Transaction ID: 0xbb05
Response status: No error
Answer: 142.250.201.238
Record type: A
TTL in captured answer: 14 seconds
Response time (Wireshark): 3.319700 milliseconds
Also visible in screenshot: Frame length 95 bytes; IP TTL 64; flags 0x8180 (standard query response, no error); 1 answer RR; “Request In: 236”

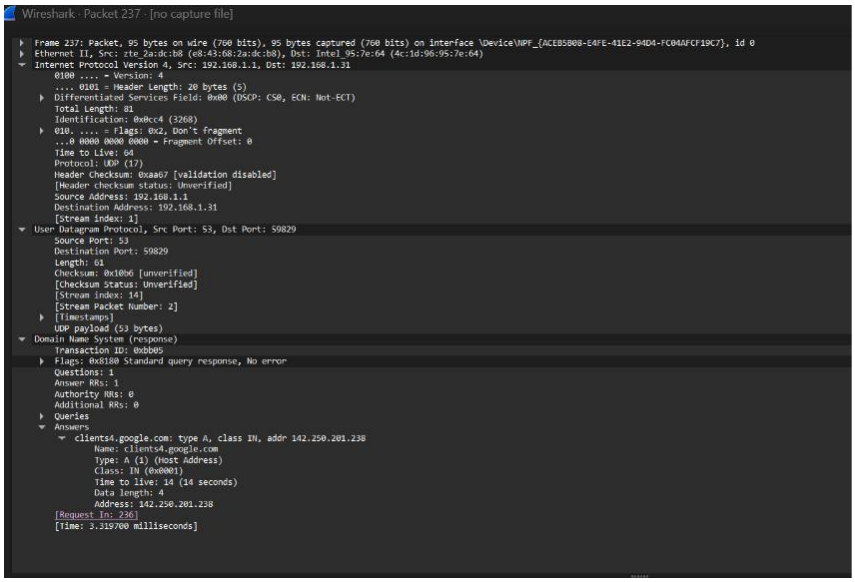


Figure 4 – *Expanded response packet 237 showing the A record answer (142.250.201.238), TTL of 14 seconds and a response time of 3.319700 ms.*

5.4 Query–Response Correlation

Field	Query (packet 236)	Response (packet 237)
Source IPv4	192.168.1.31	192.168.1.1
Destination IPv4	192.168.1.1	192.168.1.31
UDP source port	59829	53
UDP destination port	53	59829
Transaction ID	0xbb05	0xbb05
Name / type	clients4.google.com / A	clients4.google.com / A

Field	Query (packet 236)	Response (packet 237)
Result	–	No error; 142.250.201.238

The matching transaction ID links the response to the original query. The source and destination addresses and UDP ports reverse in the response. The A record provides an IPv4 address for the requested hostname.

The TTL value indicates the remaining time, in seconds, that this DNS answer may be cached according to the record's advertised lifetime. The observed value was 14 seconds at the time of capture.

6. DNS Record Types Investigated

The following record types were queried and observed in Wireshark.

Type	Query	Observed result	Purpose
A	google.com	142.250.201.110	Maps a hostname to an IPv4 address.
AAAA	google.com	2a00:1450:4019:810::200e	Maps a hostname to an IPv6 address.
MX	google.com	smtp.google.com (priority 10)	Identifies mail servers responsible for receiving email for a domain.
NS	google.com	ns1.google.com, ns2.google.com, ns3.google.com, ns4.google.com	Identifies the authoritative name servers for a DNS zone.
TXT	google.com	Multiple TXT answers	Stores text data used for domain verification and email-security policies.
CNAME	www.microsoft.com	www.microsoft.com-c-3.edgekey.net	Identifies an alias that points to another hostname.
PTR	Reverse lookup (packets 31/32)	gpon.net	Maps an IP address to a hostname through reverse DNS.

6.1 A – Address Record

A record – google.com

Observed IPv4 answer: 142.250.201.110

Purpose: Maps a hostname to an IPv4 address.

Packet: Response 34 (request 33)

Transport (screenshot): IPv6: fe80::1 → fe80::411f:8ea:7a1e:3000; UDP 53 → 63181

Transaction ID / status: 0x0002 / No error

TTL / response time: 190 seconds / 6.797400 ms

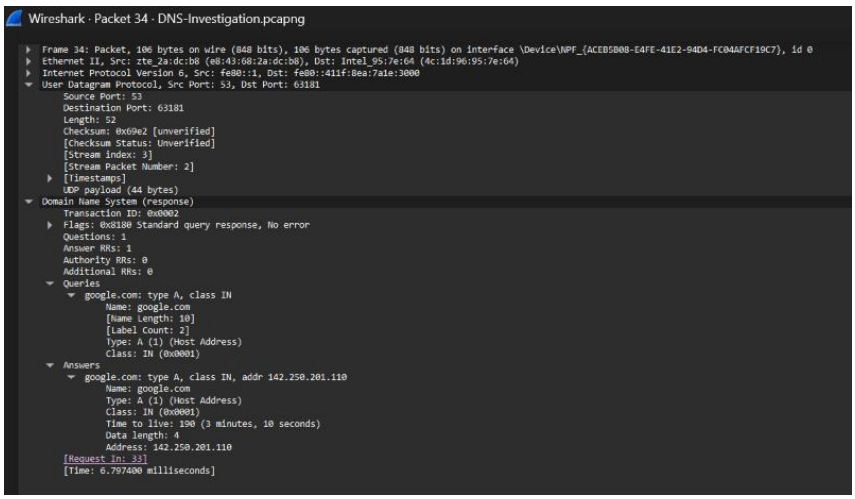


Figure 5 – *Packet 34 – A record response for google.com returning 142.250.201.110.*

6.2 AAAA – IPv6 Address Record

AAAA record – google.com

Observed IPv6 answer: 2a00:1450:4019:810::200e

Purpose: Maps a hostname to an IPv6 address.

Packet: Response 38 (request 37)

Transport (screenshot): UDP 53 → 63183 over IPv6 link-local

Transaction ID / status: 0x0002 / No error

TTL / response time: 107 seconds / 10.796400 ms

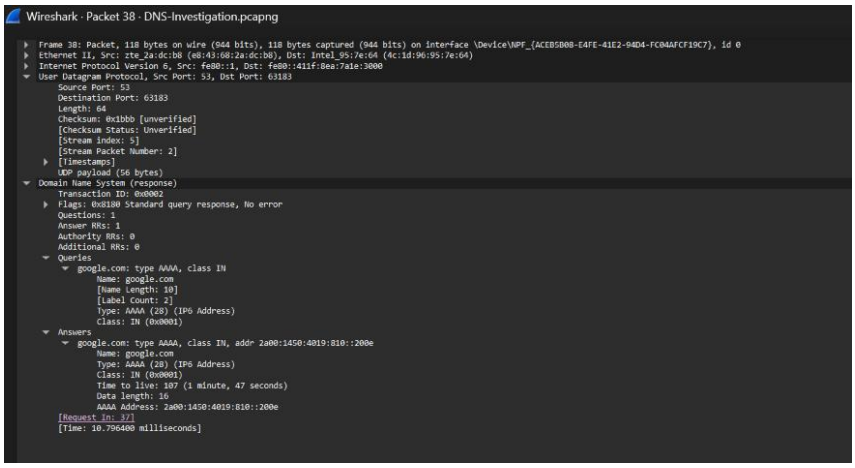


Figure 6 – *Packet 38 – AAAA record response for google.com returning 2a00:1450:4019:810::200e.*

6.3 MX – Mail Exchange Record

MX record – google.com

Observed mail exchanger: smtp.google.com

Observed priority: 10

Purpose: Identifies mail servers responsible for receiving email for a domain.

Packet: Response 42 (request 41)

TTL / response time: 300 seconds / 174.350100 ms

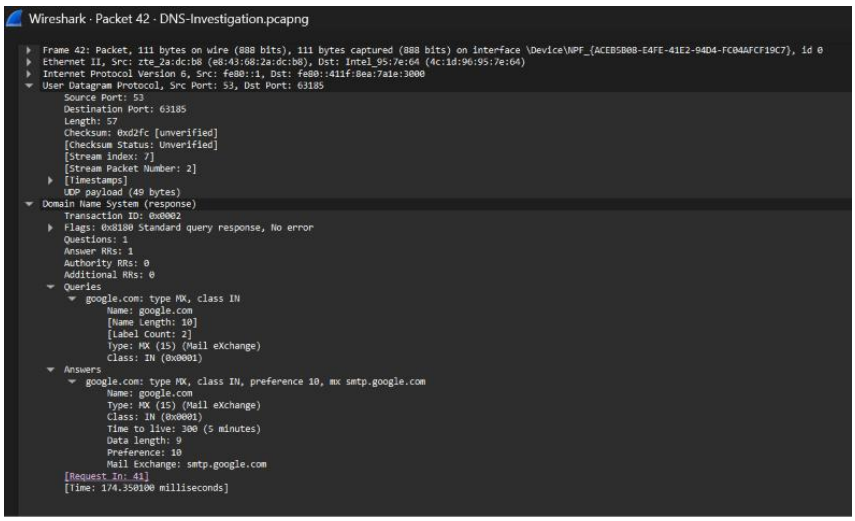


Figure 7 – *Packet 42 – MX record response for google.com: preference 10, mail exchanger smtp.google.com.*

6.4 NS – Name Server Record

NS record – google.com

Observed name servers: ns1.google.com, ns2.google.com, ns3.google.com, ns4.google.com

Purpose: Identifies the authoritative name servers for a DNS zone.

Packet: Response 46

Also visible in screenshot: Four NS answers (TTL 170149 s) plus additional A records for the name servers, e.g. ns1.google.com 216.239.32.10 and ns3.google.com 216.239.36.10 (view cropped)

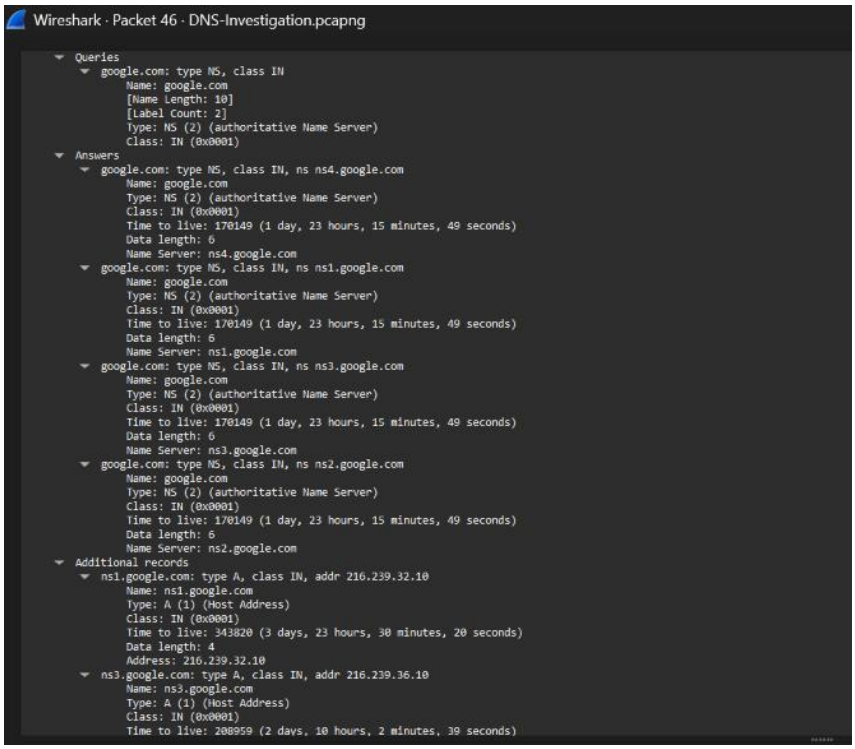


Figure 8 – *Packet 46 – NS record response for google.com listing ns1–ns4.google.com with additional records.*

6.5 TXT – Text Record

TXT record – google.com

Observation: Wireshark displayed multiple TXT answers.

Purpose: Stores text data used for purposes such as domain verification and email-security policies.

Packet: Response 51

Note: The capture summary confirmed multiple TXT records; individual TXT values were not transcribed in this investigation. The screenshot (cropped) shows several answers with a TTL of 300 seconds.

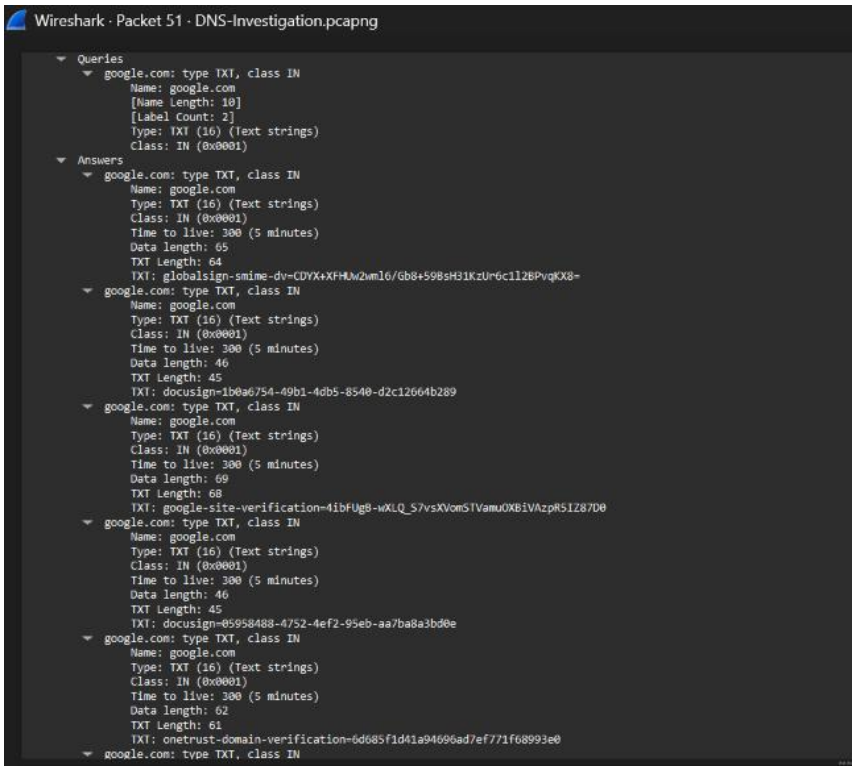


Figure 9 – *Packet 51 – TXT record response for google.com showing multiple TXT answers.*

6.6 CNAME – Canonical Name Record

CNAME record – www.microsoft.com

Observed canonical target: www.microsoft.com-c-3.edgekey.net

Purpose: Identifies an alias that points to another hostname.

Packet: Response 289 (request 288)

Transaction ID / status: 0x0002 / No error

TTL / response time: 435 seconds / 5.872600 ms

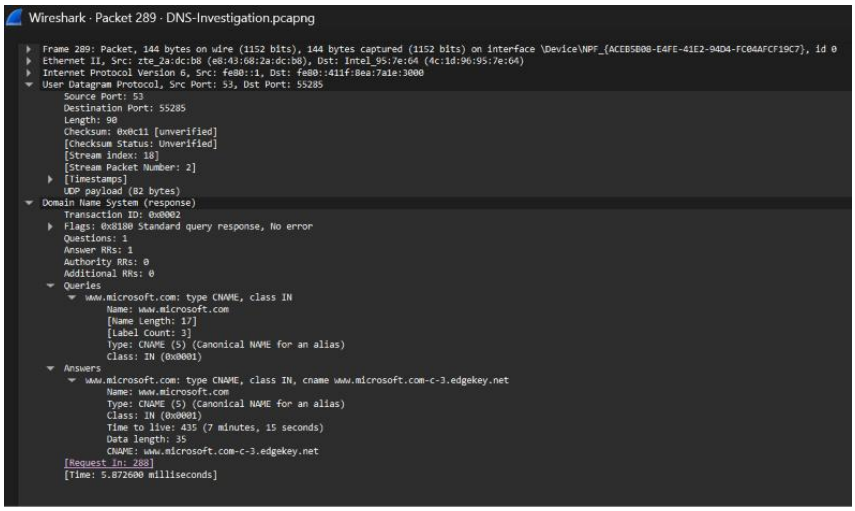


Figure 10 – *Packet 289 – CNAME response for www.microsoft.com pointing to www.microsoft.com-c-3.edgekey.net.*

6.7 PTR – Pointer Record (Reverse DNS)

PTR record – resolver lookup

Query packet: 31

Response packet: 32

Result: The reverse lookup response returned the hostname gpon.net for the resolver's IPv6 link-local address.

Purpose: Maps an IP address to a hostname through reverse DNS.

Also visible in screenshot: Queried name 1.0.0.0…8.e.f.ip6.arpa (the fe80::1 reverse name); status No error; TTL 0 s; response time 9.484900 ms

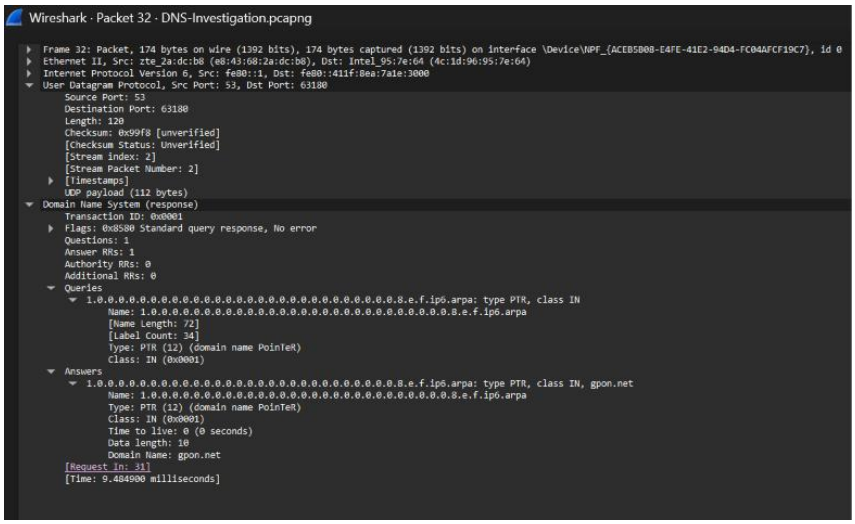


Figure 11 – *Packet 32 – PTR response returning gpon.net, confirming the resolver hostname.*

7. Additional Observations

- Both IPv4 and IPv6 DNS traffic were present in the capture.
- DNS requests and responses were primarily observed over UDP port 53.
- A PTR lookup for the resolver hostname was visible in the capture.
- The capture included an HTTPS-type DNS query for chat.openai.com that received a Refused response, while an A query for the same hostname received an answer. This was recorded as a resolver response/policy observation; it is not, by itself, evidence of malicious activity.
- DNS answers observed during separate lookups were not always identical (for example, google.com resolved to 142.250.200.174 in nslookup and 142.250.201.110 in the A-record capture). DNS responses can change because of caching, load balancing, geographic routing, and changes in service infrastructure.

8. Findings

The investigation successfully captured and analyzed real DNS traffic. Multiple domains were resolved, the local DNS resolver was identified, and DNS query/response pairs were examined at packet level.

1. The local router/resolver at 192.168.1.1 (fe80::1 over IPv6, hostname gpon.net) provides DNS service to the client at 192.168.1.31.
2. Transaction IDs associate queries with responses: query 236 and response 237 share ID 0xbb05.
3. Conventional DNS traffic uses UDP port 53 on the server side, with the source and destination addresses and ports reversed in the response.
4. The investigation identified A, AAAA, MX, NS, TXT, CNAME and PTR records, each providing a different kind of information.
5. A DNS response status (No error), an answer TTL (14 seconds for packet 237) and a measured query–response interval (3.319700 ms) were documented.
6. No conclusion of malicious activity was made from the observed traffic.

9. Conclusion

This practical investigation demonstrated the DNS resolution process through live packet capture and analysis. Using Wireshark and nslookup, DNS traffic was generated, filtered, and examined to identify the client, resolver, transport protocol, ports, transaction IDs, response status, answer data and TTL.

The work provided practical experience interpreting DNS records and understanding the role DNS plays in normal network communication.

10. Evidence and Artifacts

The packet capture was saved separately as DNS-Investigation.pcapng. The screenshots included in this report cover:

- Successful nslookup results for multiple domains
- Wireshark DNS packet capture and display-filter results
- Expanded DNS query and response packets (236 and 237)
- DNS answer details, including the A record and TTL
- A, AAAA, MX, NS, TXT and CNAME record observations
- PTR query and response identifying gpon.net

List of figures

Figure	Evidence	Location in report
1	nslookup results (cache flush + three lookups)	Section 3
2	Wireshark packet list, dns display filter	Section 5.1
3	Query packet 236	Section 5.2
4	Response packet 237	Section 5.3

Figure	Evidence	Location in report
5	A record response – packet 34	Section 6.1
6	AAAA record response – packet 38	Section 6.2
7	MX record response – packet 42	Section 6.3
8	NS record response – packet 46	Section 6.4
9	TXT record response – packet 51	Section 6.5
10	CNAME response – packet 289	Section 6.6
11	PTR response – packet 32	Section 6.7

End of report.